

NGS INTERNSHIP · MINI PROJECT

Variant Calling and Annotation

Presented by Team 7

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What is Variant Calling?

Variant calling identifies differences between sequenced DNA and a reference genome, revealing SNPs, Insertions/Deletions (InDels), and Structural Variations (SVs) across the entire genome.

SNP

Single base substitution
(e.g. A → G)

InDel

Insertion or deletion
of 1–50 bases

SV

Large structural
rearrangements (>50 bp)

Why Variant Calling?

01

Gene Fine Mapping

Provides the data foundation for subsequent functional gene fine mapping studies

02

Genome Comparison

Quickly, accurately and efficiently analyzes differences between genomes

03

Base-Level Analysis

Analyzes every single base across the whole genome systematically

04

Molecular Markers

Obtains the most extensive set of molecular markers for research

05

Disease Mutations

Identifies potential disease-causing mutations for clinical and research applications

Key Challenges

Noise vs. Real Mutations

Distinguishing true variants from sequencing errors, PCR artifacts, and alignment noise is the core challenge of the entire pipeline.

Allelic Drop-out

Preferential PCR amplification of one allele creates false allele frequency imbalances that can mimic somatic mutations.

Reference Bias

Reads with alternative alleles may align less well, causing systematic under-calling of heterozygous variants.

Computational Scale

WGS samples are 40–50 GB each. Processing requires 100+ GB RAM and multi-day runtimes on high-memory servers.

Dataset & Sample Information

BioProject: PRJEB62494 · Whole Genome Sequencing of Tongue Cancer Samples & Cell Line

3

WGS Samples

Paired-end Illumina HiSeq 1500

150 bp

Read Length

Sanger / Illumina 1.9 encoding

44%

GC Content

~1M reads downloaded per sample

chr20 + 21

Target

hg38 reference chromosomes

SRA Accessions

Accession ID	Spots	Size	GC%
ERR11468775	410.1M	46.3 GB	44.0%
ERR11468776	408.4M	46.2 GB	43.8%
ERR11468777	394.6M	46.6 GB	44.1%

Methods

Alignment-based methods: map sequencing reads to a reference genome and identify variants based on differences between the reads and the reference.

BWA-MEM

Burrows-Wheeler
Aligner
(Maximal Exact Matches)
Fast read alignment to
reference

GATK

Genome Analysis Toolkit
Best-practices variant
calling,
BQSR, Mutect2,
HaplotypeCaller

SAMtools

SAM/BAM file
manipulation,
sorting, indexing,
flagstat,
duplicate marking
support

Ensembl VEP

Variant Effect Predictor
Annotates functional
effects:
codon & amino acid
changes

Pipeline for Variant Calling

End-to-end GATK Best Practices workflow

4 steps

Pre-processing

QC → Alignment → BAM

3 steps

Variant Calling

Somatic + Germline

2 steps

Annotation

VEP + Filtering

Variant Calling Pipeline

PRE-PROCESSING

Raw Reads (FASTQ)

ERR11468775/776/777 · 1M reads

Quality Check — FastQC

Per-base quality · Adapter content

Adapter Trim — fastp

-f 10 · -l 50 · -g · -c correction

Align — BWA-MEM

Map to chr20 & chr21 (hg38)

SAM → BAM Conversion

samtools view -bo

Add Read Groups — GATK

AddOrReplaceReadGroups

Coordinate Sort

GATK SortSam -SO coordinate

GATK PROCESSING

Mark PCR Duplicates

GATK MarkDuplicates · metrics file

Base Recalibration

BaseRecalibrator + dbSNP138

Apply BQSR

ApplyBQSR → recal BAM

BAM Index

samtools index → .bai file

VARIANT CALLING

Variant Calling

GATK Mutect2 + HaplotypeCaller

Filter Variants

FilterMutectCalls · QUAL > 50

Extract SNPs & Indels

bcftools view --types snps/indels

Filter SNPs / Indels

Apply quality thresholds

Predict Variant Effects

Ensembl VEP · GRCh37/38

Annotated VCF / HTML

VCF · TXT · HTML outputs

Results – Quality Control & Trimming

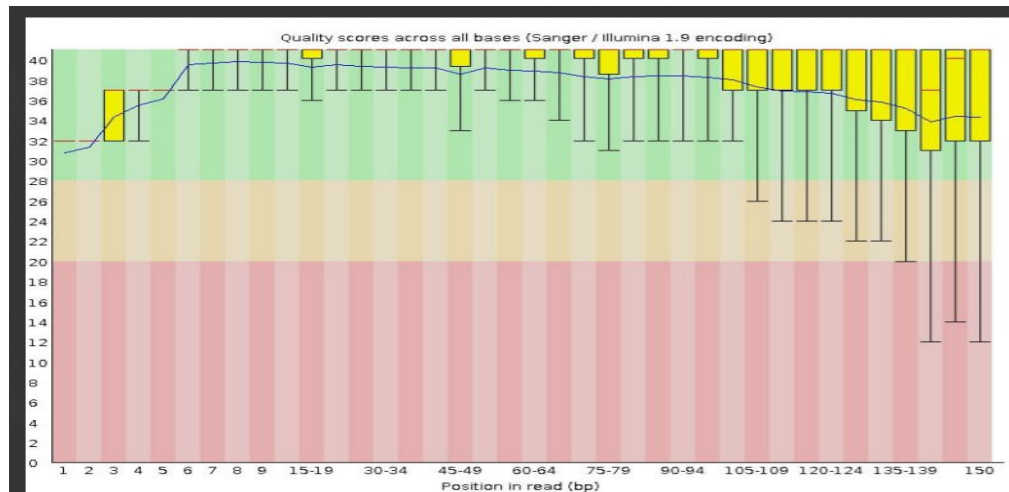
FastQC — Basic Statistics



Basic Statistics

Measure	Value
Filename	ERR11468777_2.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1000000
Sequences flagged as poor quality	0
Sequence length	150
%GC	44

Quality Scores Across All Bases



fastp Filtering Summary

Reads passed filter: **1,952,106**

Reads failed (low quality): **45,228**

Reads with adapter trimmed: **149,528**

Duplication rate: **4.03%**

Insert size peak: **216 bp**

Reads corrected by overlap: **139,002**

Results — Variant Calling

1 Call Variants (HaplotypeCaller)

- GATK HaplotypeCaller maximises sensitivity
- Minimises false negatives — calls both SNPs & indels
- Output: single VCF per sample per chromosome
- SNPs and indels extracted to separate files

2 Filter Variants

- First pass is highly sensitive — many false positives
- FilterMutectCalls applies ML artifact model
- QUAL > 50 threshold retains high-confidence calls
- Highest-quality call set passed to annotation

SNP variants per sample — Chromosome 20 results

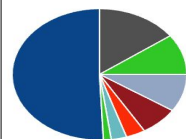
Variant Effect Predictor results ?

[Job details](#) ⓘ

[Summary statistics](#) ⓘ

Category	Count
Variants processed	41
Variants filtered out	0
Novel / existing variants	39 (95.1) / 2 (4.9)
Overlapped genes	10
Overlapped transcripts	40
Overlapped regulatory features	3

Consequences (all)



intron_variant: 51%
intergenic_variant: 15%
non_coding_transcript_variant: 10%
downstream_gene_variant: 9%
regulatory_region_variant: 7%
NMD_transcript_variant: 4%
3_prime_UTR_variant: 3%
non_coding_transcript_exon_variant: 1%

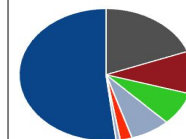
Variant Effect Predictor results ?

[Job details](#) ⓘ

[Summary statistics](#) ⓘ

Category	Count
Variants processed	47
Variants filtered out	0
Novel / existing variants	44 (93.6) / 3 (6.4)
Overlapped genes	12
Overlapped transcripts	45
Overlapped regulatory features	5

Consequences (all)

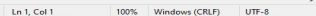


intron_variant: 52%
intergenic_variant: 19%
regulatory_region_variant: 10%
non_coding_transcript_variant: 8%
downstream_gene_variant: 7%
NMD_transcript_variant: 2%
upstream_gene_variant: 1%

Chromosome 20 — Sample 1, 2 and 3

Chromosome 20 — Combined view

Chromosome 21



Results — Variant Annotation (VEP)

Ensembl VEP — GRCh37/38

- Codon changes
- Amino acid changes
- Genomic region classification
- Functional effect: silent / missense
- SIFT & PolyPhen predictions
- HIGH / MODERATE / LOW impact tiers

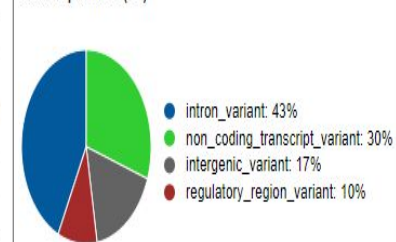
Variant Effect Predictor results ?

Job details ?

Summary statistics ?

Category	Count
Variants processed	36
Variants filtered out	0
Novel / existing variants	35 (97.2) / 1 (2.8)
Overlapped genes	5
Overlapped transcripts	18
Overlapped regulatory features	1

Consequences (all)



Annotation output — codon changes, amino acid substitutions, variant effects

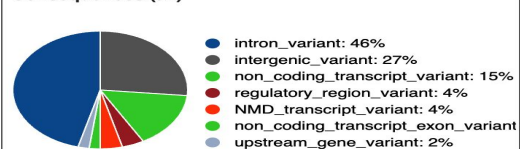
Variant Effect Predictor results ?

Job details ?

Summary statistics ?

Category	Count
Variants processed	45
Variants filtered out	0
Novel / existing variants	43 (95.6) / 2 (4.4)
Overlapped genes	8
Overlapped transcripts	23
Overlapped regulatory features	2

Consequences (all)



VEP

Web Tools

- Web Tools
- BLAST/BLAT
- Variant Effect Predictor
- VEP analysis of pasted data
- Linkage Disequilibrium Calculator
- Variant Recorder
- File Chameleon
- Assembly Converter
- ID History Converter
- VCF to PED Converter
- Allele Frequency Calculator
- Data Slicer
- Variation Pattern Finder

Configure this page

Custom tracks

Export data

Share this page

Bookmark this page

Variant Effect Predictor results ?

Job details ?

Summary statistics ?

Category	Count
Variants processed	39
Variants filtered out	0
Novel / existing variants	37 (94.9) / 2 (5.1)
Overlapped genes	5
Overlapped transcripts	17
Overlapped regulatory features	0

Consequences (all)



Results preview

Navigation (per variant)

Page: 1 of 0 variants

Show: 10 variants

Filters

Uploaded variant

is

defined

Add

Download

All: VCF VEP TXT

BioMart: Variants Genes

New job

Uploaded variant	Location	Allele	Consequence	Symbol	Gene	Feature type	Feature	Biotype	Exon	cDNA position	REF	UPLOADED
rs145874	1	A	missense_variant	SLC6A4	SLC6A4	Transcript	SLC6A4	Transcript	1	1	A	1

Acknowledgement

We sincerely thank our supervisors for their guidance and support throughout this project.

NyBerMan Bioinformatics Team

Internship coordination & mentorship

Dr. Sreeram Peela

Project supervision & technical guidance



Thank You

Variant Calling and Annotation · NGS Internship Mini-Project · Team 7

